capture of neutrons, as by the $_{92}U^{238}$ nucleus, is a reaction the occurrence of which is more likely in the case of certain values of the neutron energy, where the incident neutron wave vibrates in resonance with the vibrations of the nucleus.

In the fission process, as a rule, a few neutrons are also hurled out from the nucleus; thus, for instance, in the fission of the $_{92}U^{235}$ nucleus, two or three neutrons are emitted. This phenomenon, first verified as a fact by Joliot in 1939, makes possible the occurrence of the chain reaction necessary for the technical exploitation of nuclear energy. When a fission reaction takes place in a sufficiently large mass of—for instance—pure $_{92}U^{235}$, the neutrons released by it collide with other $_{92}U^{235}$ nuclei, causing fission in the latter, too, with the result that still more neutrons are liberated; these, in turn, produce fission in still other nuclei, and so on, so that finally, the entire mass of the substance is changed and a vast quantity of energy is released in the process. Owing to the high velocity of the neutrons, the whole process takes place in less than one-millionth of a second. It is, therefore, quite obvious that a sufficient quantity of pure $_{92}U^{235}$ (or a sufficient quantity of pure $_{94}Pu^{239}$) is an explosive of inconceivable power. Atomic bombs are made of these substances—and their destructive power is well known. A prerequisite for the occurrence of the chain reaction is, of course, that the mass of the ‘explosive’ should be sufficiently large, for if it is too small, the neutrons would escape through the surface before causing the fission of other nuclei. Therefore, a small mass or quantity of the above-mentioned ‘explosives’ is totally harmless. However, as soon as their mass or quantity exceeds a certain magnitude, the explosion occurs immediately, and spontaneously. Therefore, the atomic explosion is started by combining small pieces of the ‘explosive’ into a big piece (by throwing them together), and the big piece thus formed explodes at once.

Oppenheimer was the scientific leader of the atomic bomb construction in America. The production of sizable quantities of the ‘explosives’ $_{92}U^{235}$ and $_{94}Pu^{239}$, calls for such a tremendous technical effort that only the vast industrial capacity of the United States of America was in the position to afford it. In Germany, their production was not attempted during the war, for the capacity of the already overburdened German industry would not have been sufficient for it.[d]

III. THE URANIUM REACTOR

The use of nuclear energy for peaceful purposes is more important than the